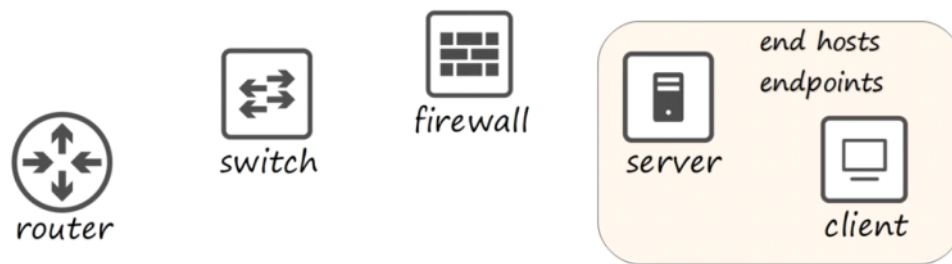


Network Devices

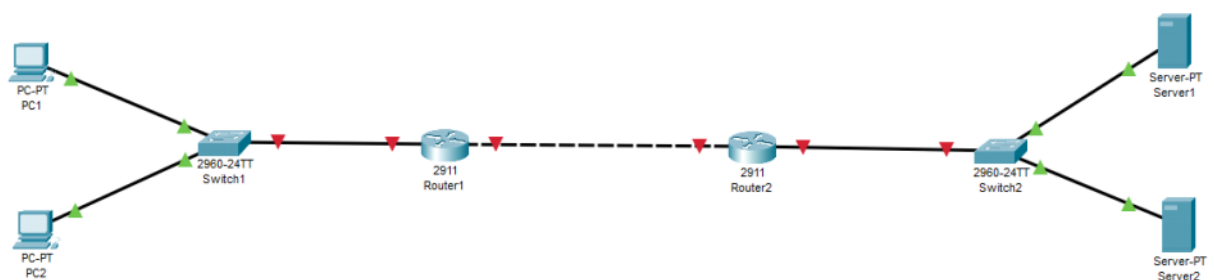
11 February 2026 09:01

Computer Network - is a digital telecommunications network which allows nodes to share resources



Network layout:

- Switch 1 is connected to PC1 and PC2.
- Switch 2 is connected to Server1 and Server2.
- Router 1 is connected to Switch 1.
- Router 2 is connected to Switch 2.
- Router 1 and Router 2 are connected to each other.



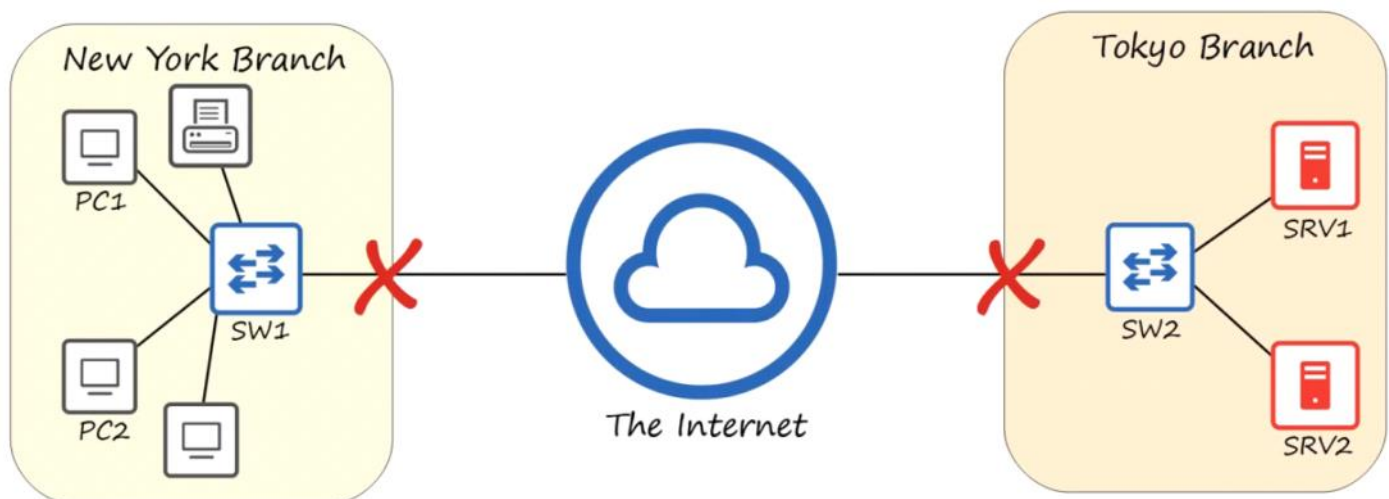
Switches

11 February 2026 09:40



LAN (Local Area Network)

A network switch is used to connect devices inside a Local Area Network (LAN) and allow them to communicate efficiently. They can share information within their branch or within the LAN (Local Area Network).



But they can't share the resources to each other branch or through the internet as switch is made to share within the LAN.



Catalyst 9200



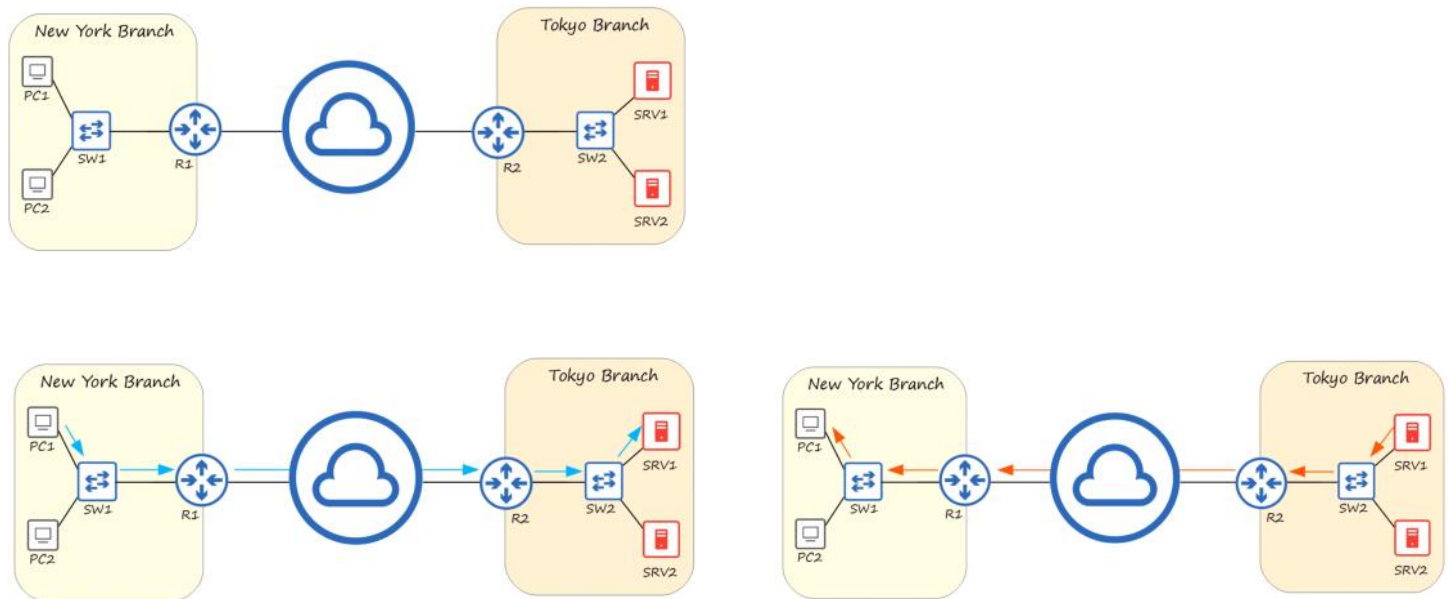
Catalyst 3650

Switches...

- have many network interfaces/ports for end hosts to connect to (usually 24+).
- provide connectivity to hosts within the same LAN (Local Area Network).
- do not provide connectivity between LANs/over the Internet.

Router

11 February 2026 18:56



ISR 1000



ISR 900



ISR 4000

ISR - Integrated Service Router



Catalyst 9200

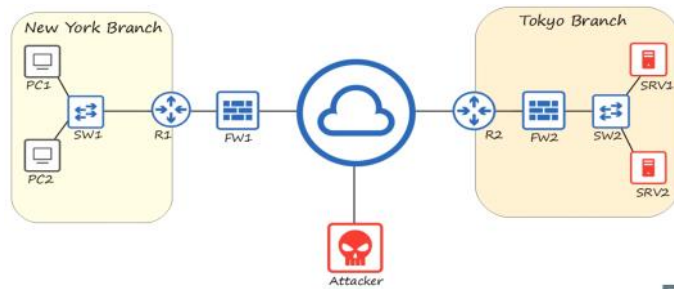
switch

Routers :

- > have fewer network interfaces than switches
- > are used to provide connectivity between LANs
- > are therefore used to send data over the internet

Firewall

11 February 2026 19:25



ASA5500-X



Firepower 2100

Firewalls...

- monitor and control network traffic based on configured rules.
- can be placed 'inside' the network, or 'outside the network'
- are known as 'Next-Generation Firewalls' when they include more modern and advanced filtering capabilities.

Network firewalls...

are hardware devices that filter traffic between networks.

Host based firewall - are installed in your device or computer
- they work as a secondary layer of protection

Host-based firewalls...

are software applications that filter traffic entering and exiting a host machine, like a PC.

MCQs

12 February 2026 08:42

Your company wants to purchase some network hardware to which they can plug the 30 PCs in your department. Which type of network device is appropriate?

- a) A router
- b) A firewall
- c) A switch
- d) A server

You received a video file from your friend's Apple iPhone using AirDrop. What was his iPhone functioning as in that transaction?

- a) A server
- b) A client
- c) A local area network

What is your computer or smartphone functioning as while you watch this video?

- a) A server
- b) An end host
- c) A client

Your company wants to purchase some network hardware to connect its separate networks together. What kind of network device is appropriate?

- a) A firewall
- b) A host
- c) A LAN
- d) A router

Your company wants to upgrade its old network firewall that has been in use for several years to one that provides more advanced functions. What kind of firewall should they purchase?

- a) A host-based firewall
- b) A next-level firewall
- c) A next-generation firewall
- d) A top-layer firewall

MCQs Answer

12 February 2026 09:09

A Switch

A server

A client

A router

A next generation firewall

Packet Tracer

12 February 2026 09:08

